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Secure and trusted data sharing in smart healthcare using blockchain and IoT integration

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Abstract

Remote service delivery and automation using Blockchain and Internet of Things (IoT) are revolutionising healthcare operations. Enhancing healthcare data interchange and providing real-time treatment is becoming more difficult due to the exponential growth of patient populations around the globe. It is still very difficult to develop a digital healthcare platform that is entirely decentralised, secure, trustworthy, interoperable, and scalable, even if existing studies have improved these platforms to improve patient outcomes and reduce hospital visits. This paper proposes a robust and scalable healthcare architecture using Blockchain smart contracts and IoT, with the integration of the InterPlanetary File System (IPFS). By securely storing sensitive medical records, the framework enhances data privacy and interoperability for patients as well as healthcare professionals. A device proxy monitors potentially vulnerable IoT devices and uses cryptography to ensure that data remains private. Experimental evaluations of the system's performance have focused on key factors, including healthcare record upload, download, access, and mining times. The results show that public healthcare systems based on the Blockchain considerably boost efficiency and performance by integrating IPFS.

Keywords Blockchain, IoT, IPFS, Scalability, Smart healthcare, Security, Proxy device

1 Introduction

With the growth in the global population, the number of patients are gradually rising. Healthcare organizations are advancing and adopting the latest technologies, which alter how this generation is treated. The most exciting technological advancement is the Internet of Things (IoT), which connects things and enables them to make decisions independently in intelligent environments. In terms of offering a framework for tying together everyday smart devices to collect, share, transmit, and manage data, IoT has performed better than the expectations made by researchers during the past decades. IoT technology connects heterogeneous healthcare sensor devices to improve patient care by minimizing hospital visits [1, 2]. Due to the proliferation of IoT medical devices, healthcare organizations must regularly gather, store, process, and provide access to enormous amounts of healthcare data. Centralized storage architecture in most IoT healthcare systems can lead to a single point of failure and easy manipulation.



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Unfortunately, privacy and security of personal health information are less reliably protected in IoT healthcare systems [3, 4]. Some inexpensive IoT devices used in IoT systems are very easy targets for malicious nodes. Due to IoT infrastructure storage and memory limits, the healthcare system occasionally stops working when a vast volume of healthcare data is collected from many linked sensor devices [5]. Despite cloud, edge, fog, and mist computing systems offering almost endless storage and processing capacity, healthcare data exchanges' traffic, bandwidth, and processing latency cause the network bottleneck. IoT-based healthcare technology faces substantial performance, reliability, scalability, interoperability, privacy, trust, decentralization, and security challenges [6]. Blockchain with IoT integration can help healthcare systems overcome centralized communication architecture and cloud-enabled IoT infrastructures. Most healthcare organizations adopt Blockchain, a modern digital ledger technology (DLT), to solve technical issues when creating IoT healthcare systems. Blockchain's peer-to-peer network performs cryptographically hashed, immutable transactions on a decentralized architecture [7, 8]. Blockchain technology allows data transfers without a central authority, eliminating a single point of failure in IoT healthcare systems. Due to Blockchain technology's distributed computation and decentralized storage, large-scale device connectivity, massive data collection, processing, control, communication, storage, and analysis are now more feasible than in conventional healthcare systems [9, 10]. Blockchain technology allows doctors, patients, diagnostic centres, and healthcare workers to deliver remote patient care, lower treatment costs, and improve patient care. There are three distinct types of Blockchains: public, private, and consortium. Blockchains have different access mechanisms and verification procedures, which help to strengthen the Blockchain-enabled healthcare system. Proof of work is used to verify the creation of new blocks in Blockchain technology [5, 11]. Ethereum smart contract automatically executes healthcare regulations, policies, and the device registration process when large-scale healthcare devices are connected to the system. The third-party verification process is eliminated with this process. With Blockchain-integrated IoT healthcare systems, data tampering, malicious activity, and unsecured devices can be easily identified and traced. Blockchain-based healthcare solutions simplify the management of current and historical medical records through its decentralized nature. Blockchain technology guarantees the confidentiality, security, dependability, and integrity of sensitive healthcare information [12, 13]. While Blockchain healthcare has several advantages over conventional healthcare systems, the growing complexity of managing an increasing number of devices and their associated data makes it more time-consuming to access, process, and analyze this information [14]. The Interplanetary File System (IPFS), which stores data whose hash value is accessible on Blockchain networks, can be used to reduce the overload in Blockchain networks. A decentralized, secure, and irreversible healthcare platform has been the primary focus of most earlier research. The Blockchain network has scalability, speed, and security issues when dealing with massive amounts of data collected, processed, and stored using large-scale devices. Therefore, IPFS solves this problem and makes a scalable healthcare platform. To store the massive amounts of healthcare data generated by networked sensor devices, a Blockchain network is coupled with the decentralized file-sharing platform IPFS. To obtain the data we need, IPFS uses content addressing, which operates on a peer-to-peer network [15]. When a file is uploaded to IPFS, it is broken down into smaller pieces called chunks, and these

chunks are then distributedly stored with their unique cryptographic hash, known as the content identifier (CID), attached to them. CID links all the connected hashes to store the root hash, allowing the information chunks to be uniquely identified by their hash values. An open network like BitTorrent allows for the movement of files. In IPFS, the addresses of stored data are maintained in a DHT. IPFS uses a Merkle Tree structure to store the hash values of linked files. To improve healthcare systems' speed, accessibility, storage capacity, and scalability, blockchain networks only save the file address rather than the entire file. The Blockchain network within healthcare organizations are linked to the IPFS similarly to how webpages are linked. IPFS keeps a local hash file for frequently viewed files and enables rapid access to these files on subsequent tries. Anyone accessing the hash address can download a file via the IPFS distributed network [16]. In IPFS, once a file is uploaded, its hash is shared among all peer nodes in the network for instant access and to prevent data from having a single point of failure. Here, scalability, interoperability, and consistency issues are addressed with the IPFS, which also offers security and boosts the efficiency of healthcare systems.

1.1 Motivation

Although we cannot entirely eradicate every disease that affects people today, we can adopt innovative technology to reduce the severity of patients' conditions and offer superior healthcare for a healthier world. Most past research focused on sensor-based healthcare systems to deliver remote healthcare services, improve patient cure rates, and eliminate the need for patients to travel to healthcare facilities. Still, the system's privacy, scalability, security, and efficiency should have been addressed. In recent years, healthcare organizations have explored Blockchain integration with IoT to enable all IoT devices to interact without relying on trustworthy third parties. Blockchain technology has enabled healthcare businesses to effectively manage a growing number of distributed data transactions instead of traditional centralized ones. As more patients are added, the Blockchain healthcare system's ability to connect, store, preserve, transact, and retrieve data becomes more challenging. This is because the system saves data in the Blockchain network. This framework integrates an IPFS with the Blockchain network to address the challenges of scalability, interoperability, efficiency, and security and improve storage capability.

1.2 Novel contributions

1. It presents a Blockchain-based novel framework to address healthcare systems' data privacy, security, interoperability, scalability, and efficiency challenges.
2. It Integrates an Interplanetary File System (IPFS) with the Blockchain network to reduce blockchain network overload and enhance healthcare systems' scalability and storage capacity.
3. It employs a device proxy to keep track of vulnerable devices and a cryptographic message authentication mechanism to keep data confidential.

1.3 The article structure

This paper's remaining sections are organized as follows: Sect. 2 reviews the blockchain healthcare related works. The system model of the M-Blockchain framework is presented in Sect. 3. The proposed model's implementation is described in Sect. 4, whereas

Sect. 5 discusses the evaluation, outcomes, and performance analysis of the experiments conducted to test the proposed framework. Section 6 presents the findings and compares the proposed framework with the existing works. The conclusion is presented in Sect. 7.

2 Related works

This section presents the literature on the opportunities and challenges of implementing technological advancements in Smart healthcare systems. Some of the best research works published in medical fields are highlighted here.

Zheng et al. [17] proposed an IPFS-based Blockchain system to reduce the data storage in Blockchain networks and increase processing speed. They implemented this scheme on the Bitcoin Blockchain. A zigzag-based novel storage model is proposed by Chen et al. [18] to increase the block storage model. High throughput is achieved in this model with the implementation of IPFS. Pan et al. [19] proposed an edge-IoT framework, EdgeChain, based on Blockchain and smart contracts for secure and scalable communication. Jolfaei et al. [9] discuss different system models and how to improve the scalability of Blockchain-based systems. Off-chain and on-chain-based models can be used to overcome the scalability issues in Blockchain-based IoMT systems. Providing a clear route for tracing the activities and achieving a high level of trustworthiness, Nyalety et al. [20] proposed block IPFS a Blockchain-enabled IPFS. For secure storage, safe data sharing, and maintaining security while working in the system, Li et al. [21] proposed a hyperledger fabric and Interplanetary File System to safely share healthcare information. On-chain and off-chain Blockchain-based Interplanetary File system solves scalability, confidentiality, and storage problems. This model increases the confidentiality of sharing healthcare information proposed by Mani et al. [22]. A Blockchain-based Interplanetary File System is proposed by Mamun et al. [23] for implementing Blockchain EMR and providing access rules. An immutable Blockchain-based Interplanetary File System is proposed by Sultana et al. [24] for enhancing the security of healthcare data transactions. For easy access, consistency, integrity, availability, and security, Kumar et al. [25] proposed off-chain data storage. Interplanetary File System uses blockchain technology in its model. Alrebdi et al. [26] proposed a Blockchain smart contract-based decentralized medical record system with several features: immutability, anonymity, security, decentralized, and tamper-proof. Interplanetary File Systems and cloud computing technology are used here for medical record-keeping. A security and privacy protection scheme using Blockchain technology is proposed by Tan et al. [27] with traceable and direct revocation for COVID-19 records. Cipher text policy attribute-based encryption is used here. Distributed patient-centric image management (PCIM) system, an Ethereum Blockchain, and an Interplanetary File System (IPFS) are proposed by Jabarulla et al. [28] for decentralized storage and global access to medical images. A systematic review of Blockchain-based PHRs has been done by Fang et al. [11] for limitations, design choices, and future direction. The attribute-based encryption scheme is proposed by Sun et al. [29] for the storage and sharing of medical records using Blockchain technology and the Interplanetary File System.

3 System model

Processing real-time healthcare data, timely patient monitoring, and providing healthcare services are challenging for healthcare providers worldwide [30, 31]. Although most existing Blockchain-integrated IoT systems provide smart healthcare systems, as the number of connected healthcare devices grows, the system's access to, processing, and storage of healthcare data slows down. The inability to deliver continuous, real-time patient care is a major concern for many healthcare facilities. Patients may get worse or possibly die if they do not receive the best possible care immediately. To address the shortcomings of current Blockchain-integrated IoT healthcare systems, we have proposed an IPFS-integrated Blockchain-IoT healthcare system. Figure 1. depicts the proposed framework, which is secure, interoperable, and scalable for healthcare organizations. It integrates Blockchain, IoT, and IPFS (Table 1).

3.1 Framework description

In this system, real-time patient care is prioritized above all else. Still, strong privacy, security, and scalability solutions for accessing, processing, transferring, and storing healthcare data are also paramount. Each "block" in the Blockchain healthcare system is a timestamped, cryptographically linked, distributed, and immutable set of records. The proposed system employs smart contracts built on the Ethereum platform to check if IoT devices adhere to pre-defined privacy policies and other limitations. Device access permissions are granted to devices via a smart contract-defined access management and authentication method. All healthcare IoT devices used in this framework fall into secured and unsecured categories. Unsecured IoT devices are cheap and vulnerable and cannot execute privacy policies and agreements. These devices are connected via a device proxy. The device proxy authorises these gadgets to the system and monitors their behaviour. The proposed framework also uses decentralized and cryptographically

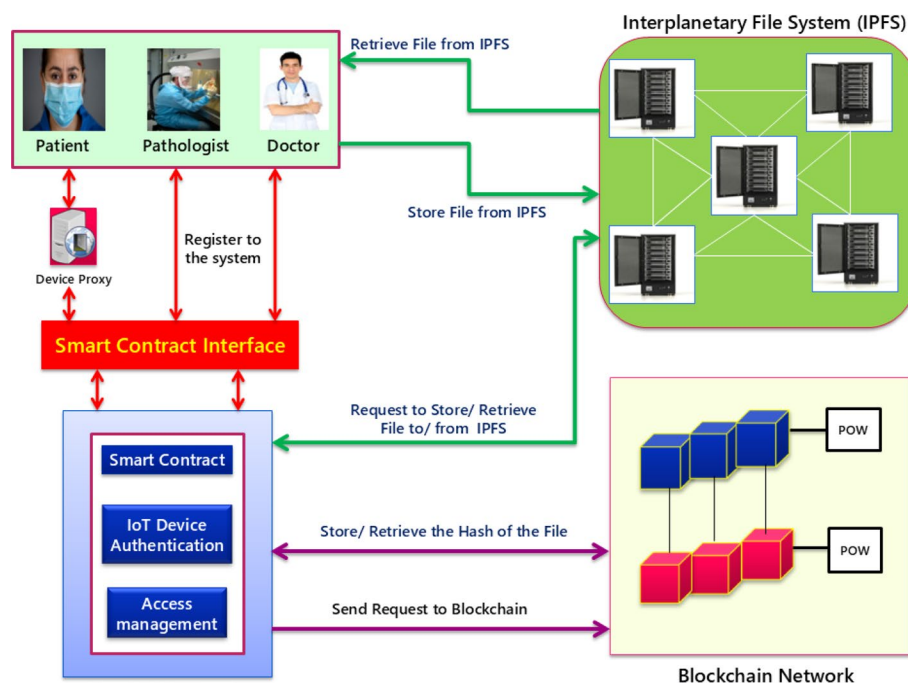


Fig. 1 Proposed healthcare framework using IPFS storage system

Table 1 Table of notations

Notations	Descriptions	Notations	Descriptions
UID	UserID	μ_{ut}	Mean of upload time
UName	User name	N_{ut}	Numbers of measured upload times
UAddress	User address	σ_{dt}	Standard deviation in download time
UContactno	User contact number	X_{dt}	Each download time count
UDob	User date of birth	μ_{dt}	Mean of download time
UIDproof	User identity proof	N_{dt}	Numbers of measured download times
Dspec	Device Specification	H_{pfile}	Hash of Patient File
IPFS	Interplanetary file system	PID	Individual patient identity
σ_{at}	Standard deviation in access time	PH_{file}	Patient health file
μ_{at}	Mean of access time	H_{file}	Hash value of requested file
N_{at}	Number of measured access times	H_{BC}	Hash value stored in Blockchain
H_{IPFS}	Hash value stored in IPFS	TXN_i	Size of each transaction
TXN	Total number of transactions	BH_s	Size of block header
HS_i	Size of each hash	BS	Block size
TXN_n	Number of transaction per block	T_i	Execution time of each transaction
B_t	Total execution time for block	PB_{ks}	Public key of sender
PB_{kr}	Public key of receiver	PV_{ks}	Private Key Sender
PV_{kr}	Private Key Receiver	$BS_{without\ IPFS}$	Block size without IPFS
X_{at}	Each access time count	$BS_{with\ IPFS}$	Block size with IPFS
X_{ut}	Each uploads time count	HUID	Healthcare User Identity
HUName	Healthcare User Name	HData	Healthcare Data
HUContactno	Healthcare User Contact Number	PID	Patient Identity
PName	Patient Name	PData	Patient Healthcare Data
PContactno	Patient Contact Number	DigSig	Digital Signature
DcrptSig	Decrypted Signature	σ_{ut}	Standard deviation in upload time

hashed IPFS to lower the network's latency, increase security, and reduce data storage costs. Unlike cloud storage, IPFS doesn't rely on a single server, so there's no need to entrust one another and no single point of failure. Each file is stored in an IPFS, with its corresponding hash maintained in the Blockchain network. Figure 2. depicts the workflow of registering and uploading patient health records. Figure 3. depicts accessing medical records stored in IPFS.

3.2 Encryption and decryption process

The two-stage process of encryption and decryption is depicted in Fig. 4. This framework hashes User1's source text into a message digest. Message digests are in an unreadable format. The source text and the sender's private key are used to construct a digital signature through encryption and hashing. The original encrypted text, the message digest, and the digital signature are sent onto the Blockchain. The receiver (user 2) separates the source text and the encrypted message digest. Using User1's public key, decrypt the encrypted message digest and extract the message digest. The message digest is computed from the original text using the same hashing technique. After computing message digests, they are compared with one another. If the digests of the two messages are the same, then the data has not been tampered with, and if they are different, then the message has been tampered with.

3.3 Minimizing data storage and transaction processing times

In this subsection, we employ quantitative analysis to demonstrate how IPFS facilitates faster transaction processing and smaller Blockchain data sizes. Equation 1 represents

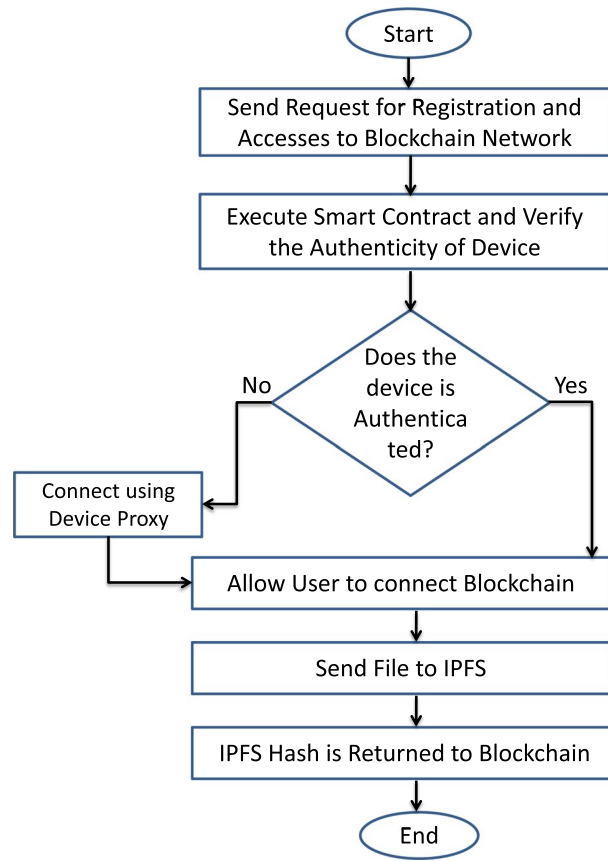


Fig. 2 Uploading healthcare data into IPFS

the reduction of block size, Eq. 2 shows the total number of transactions per block, and Eq. 3 provides the execution time of each transaction.

Block size without IPFS

$$(BS_{withoutIPFS}) = \sum_{i=1}^n TXN_i + \sum_{i=1}^n HS_i + BH_s$$

Block Size with IPFS

$$(BS_{withIPFS}) = \sum_{i=1}^n HS_i + BH_s$$

Reduction of size per block

$$\begin{aligned}
 &= (BS_{withoutIPFS}) - (BS_{withIPFS}) \\
 &= \sum_{i=1}^n (TXN_i + HS_i) + BH_s - \sum_{i=1}^n HS_i + BH_s
 \end{aligned} \tag{1}$$

Size of total transactions per block

$$TXN = BS - \sum_{i=1}^n HS_i + BH_s$$

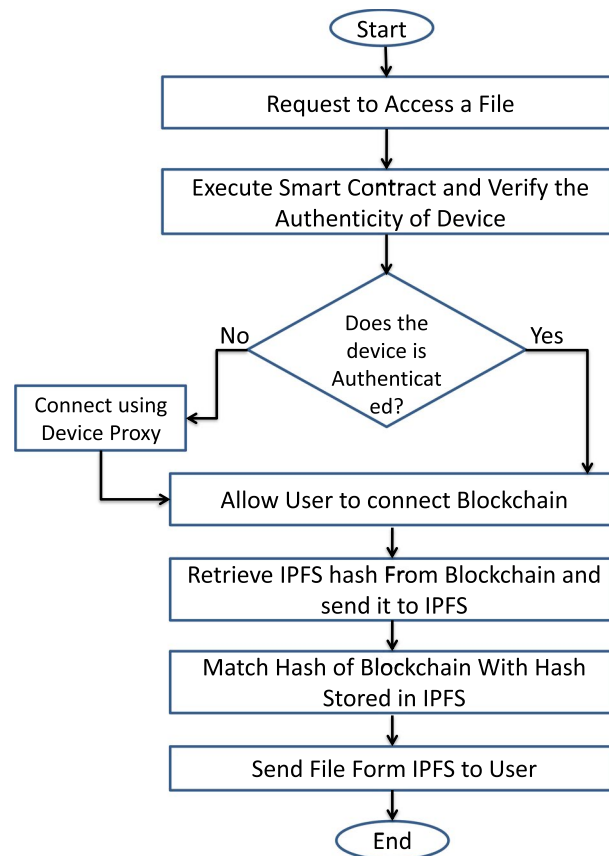


Fig. 3 Accessing healthcare data from the framework

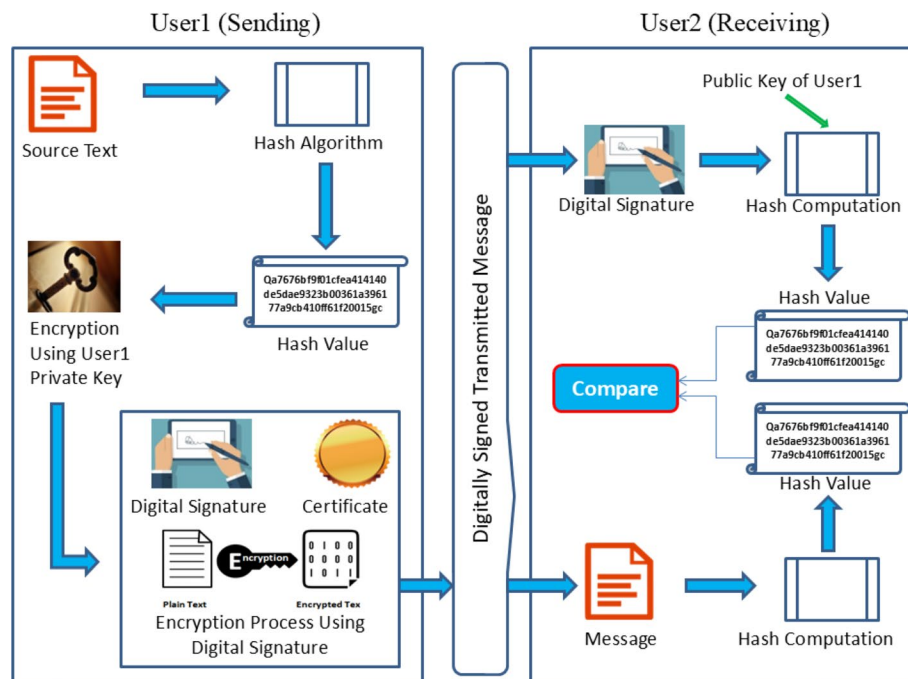


Fig. 4 Encryption and decryption process

Total number of transactions per block

$$\begin{aligned}
 TXN_n &= \frac{TXN}{\left(\frac{1}{n}\right) \sum_{i=1}^n TXN_i} \\
 &= \frac{BS - \sum_{i=1}^n HS_i + BH_s}{\left(\frac{1}{n}\right) \sum_{i=1}^n TXN_i} \quad (2)
 \end{aligned}$$

Execution time of each transaction

$$\begin{aligned}
 T_i &= \frac{\text{Total execution time for a block}}{\text{Total number of transactions}} = \frac{B_t}{TXN_n} \\
 &= \frac{B_t}{\frac{BS - \sum_{i=1}^n HS_i + BH_s}{\left(\frac{1}{n}\right) \sum_{i=1}^n TXN_i}} \\
 &= \frac{\left(\frac{1}{n}\right) (B_t \sum_{i=1}^n TXN_i)}{BS - \sum_{i=1}^n HS_i + BH_s} \quad (3)
 \end{aligned}$$

4 Implementation

4.1 Environmental setup

The proposed healthcare system was implemented in the IoT and cloud computing lab at IIIT Bhubaneswar with the system configuration of Intel Core i7-1165G7, 4.70 GHz processor, with a cache memory of 12 MB, running Windows 10 Professional on a 1 terabyte (TB) hard drive with a 512 gigabyte (GB) solid-state drive (SSD) and a 64-bit operating system. All machines are connected to the same private network for experimental evaluation. One machine is the master machine, and the others are the client machines. Different data sizes, including 20MB, 40MB, 60MB, 80MB, 100MB, 120MB, and 140MB, are chosen to evaluate the performance of the proposed model. Table 2 lists the software, frameworks, and tools used in the proposed framework.

4.2 Execution process

Three algorithms have been presented for this proposed framework. Algorithm 1 shows the registration and uploading of the file into the proposed framework. The smart contract checks device authentication when users register for the decentralized application. Some cheap gadgets today aren't secure enough to use the Blockchain network's smart

Table 2 Software/ Frameworks/ Tools used

Software's/ Frameworks/ Tools	Versions
Solidity	0.8.0
IPFS Provider	Pinata/SDK IPFS 2.1.0
Geth	1.11.16
Hash Algorithm	Keccak-256
Node.js	18.15.0
React JS	18.2.0
Truffle	5.9.1
Eslint	8.37.0
Web3.js	1.10.0
Metamask	10.30.4
Ethereum Testnet	Polygon Mumbai Testnet

contract functionalities. A device proxy is used to identify and register these devices. All actions done by these devices are watched by the device proxy. Registration of a device requires proof of identity. The decentralised application (dApp) based on smart contracts connects all users and patients to the healthcare system, addressing the concerns of security, privacy, scalability, trustworthiness, and transparency. Users' data uploaded using the decentralized app are stored in the Interplanetary File System. A cryptographic hash value generated from this file will be uploaded to the Blockchain network. A step-by-step process for using this proposed framework to retrieve the files is provided in Algorithm 2. When a user connects to the decentralized application and desires to access a file from this system, the user must first send a request to the system. The file's unique Hash value will be retrieved from the Blockchain network. This hash value will be checked against the hash values in the interplanetary file system. Once the hash value has been matched, the user will obtain the appropriate file for further processing. The encryption and decryption process of proposed healthcare system is presented in Algorithm 3.

```

1: Inputs:
   User Credentials, Patient health checkup
   report
2: Output:
   User ID generation and IPFS-based patient
   healthcare data storage
3: Procedure RegisterUser (UID, UName, UAddress,
   UContactno, UDOB, UIDproof, Timestamp, CommitAt,
   Dspec)
4: if UID exists then
5:   Alert UID already exists
6: else if UID does not exist then
7:   CreateUser  $\leftarrow$  UserCredentials {
     UID = UserIdentity,
     UName = UserName,
     UAddress = UserAddress,
     UDob = Date of Birth,
     UContactno = PhoneNumber,
     UIDproof = Valid ID Proof,
     CommitAt = Now,
     Timestamp = CurrentTimestamp,
     Dspec = Device Specification }
8: else if (IsUser == "Healthcare User" && Access ==
   "ALLOW") then
9:   SearchID  $\leftarrow$  PID
10:  Store patient files using PID to the IPFS
11:  CommitAt  $\leftarrow$  Now
12:  Timestamp  $\leftarrow$  CurrentTimestamp
13:  FileChunks  $\leftarrow$  splitFileChunks
14:  Ddata  $\leftarrow$  EncryptedData
15:  for Chunk in FileChunk do
16:    ContentHash  $\leftarrow$  IPFS Chunk Hash
17:  end for
18:  Transaction  $\leftarrow$  {
    PID,
    Content Hashes,
    Current Timestamp,
    File Uploaded}
19:  Blockchain  $\leftarrow$  Transaction
20: end if
21: endProcedure

```

Algorithm 1 Registration and Healthcare Data Upload

```

1: Inputs:
   Requested file hash
2: Output:
   Obtaining patient healthcare records
3: Procedure AccessPatientData PID, ( $P_{Hfile}$ )
4: if (Request  $H_{Pfile}$  == False) then
5:   Requested patient report not available
6: else
7:   if (Request  $H_{Pfile}$  == True) then
8:      $Transaction \leftarrow Get \quad Transaction$ 
        $ID \text{ of } (PID, P_{Hfile})$ 
9:      $ContentHash \leftarrow Transaction.ContentHash$ 
10:     $FileChunk \leftarrow Empty$ 
11:    for (ContentHash in DHT) do
12:       $FileChunk \leftarrow$ 
         $IPFS \text{ File Chunk Content Hashes}$ 
13:       $FileChunk \leftarrow Add \text{ File Chunks}$ 
14:    end for
15:     $EncryptedFile \leftarrow Combined \text{ Chunk Files}$ 
16:     $DecryptedFile \leftarrow$ 
       $DecryptedFile(EncryptedFile)$ 
17:    Return DecryptedFile
18:    Return the file, transaction success
19:  end if
20: end ifendProcedure

```

Algorithm 2 Accessing Healthcare Data From IPFS

```

1: Inputs:
   Sending Healthcare File
2: Output:
   Obtaining Healthcare File
3: Procedure DataEncryption( $PB_{Ks}$ , DataFile)
4:  $DataEncrypted \leftarrow Encryption(PB_{Ks}, DataFile)$ 
5: Return DataEncrypted
   Procedure FileHash(DataFile)
6:  $DataHash \leftarrow hash(DataFile)$ 
7: Return DataHash
8: Procedure CreateDigSig( $PV_{Ks}$ , DataFile)
9:  $DataHash \leftarrow FileHash(DataFile)$ 
10:  $DigSig \leftarrow Encryption(DataHash, PV_K)$ 
11: Return DigSig
12: Procedure ValidateDigSig( $PB_{ks}$ , DigSig, DataFile)
13:  $DataHash \leftarrow FileHash(DataFile)$ 
14:  $DcrptSig \leftarrow Decryption(DigSig, PB_{ks})$ 
15: if (DigSig==DataHash) then
16:   Return Received File is Correct
17:   Return Received File is not Correct
18: end if
   Procedure DataDecryption( $PV_{Kr}$ , DataEn-
     crypted)
19:  $DataDecrypted \leftarrow Decryption(PV_{Kr}, DataEncrypted)$ 
20: Return DataDecrypted
   endProcedure

```

Algorithm 3 Algorithm for Encrypting and Decrypting Healthcare Data

4.3 Proof of the correctness of algorithms

The use of cryptographically bound, non-transferable identification tokens, as illustrated in Algorithm 1, is integral to the system. The registration process incorporates protections to ensure the security of personal information while confirming identity. No one can deny or alter a registered identity because of the immutability of the blockchain. Consistent enforcement of registration rules across all participants is ensured by smart contracts. Each user is able to access the precise data that was initially uploaded, as shown in Algorithm 2. The data is protected so that only authorised people can access it. The integrity of the obtained data is confirmed by comparing it with IPFS hashes. Additionally, this algorithm completes its execution in a finite number of steps. Deciphering the ciphertext with the right parameters and key returns the original plaintext, as demonstrated in Algorithm 3. Hash functions on the Blockchain and IPFS CID ensure that data remains secure during storage and transaction.

5 Result analysis

We have conducted a number of experiments on various machines in the same network to evaluate the performance of our proposed framework. Using content-addressed hashing, the platform facilitates efficient distributed healthcare data storage, access, and retrieval. This architecture is user-friendly and effective at preserving the confidentiality, availability, and integrity of sensitive healthcare information. We employed seven distinct healthcare file sizes across multiple machines to test the proposed framework. Tests of the framework's performance were conducted using the metrics of upload, download, access, and mining times.

- *Upload Time*: Upload time is the time a node takes to upload new transactions to the Blockchain network. Upload time depends on network speed and node processing power.
- *Download Time*: Blockchain download time is the time it takes a node to download the file and synchronize with the whole network. Blockchain size, network speed, and node processing power determine download time.
- *Access Time*: Blockchain access time is the time it takes a user to access a transaction or block. It varies depending on the Blockchain's size and the user's device's computational power.
- *Mining Time*: Blockchain mining time is the time it takes miners to solve cryptographic puzzles and add new blocks or transactions. Puzzle difficulty and miner node processing speed affect mining time. Network protocol determines block time, which differs from each Blockchain.

We've considered healthcare files of varying sizes, uploaded them from several machines, and timed how long each file took to be added to Blockchain and IPFS. Figure 5 displays the difference in upload speed between IPFS and Blockchain. Observation indicates that the time required to upload files to the Blockchain network increases with file size. The graph in Fig. 5 indicates that IPFS uploads faster than Blockchain. The time it takes to upload a file to the IPFS network grows as the file size increases; however, after a certain file size, it increases minimally, much faster than uploading to Blockchain. Figure 6 compares Blockchain and IPFS download timings. This graph depicts the experimental model's results, which show that downloading the file through Blockchain takes noticeably

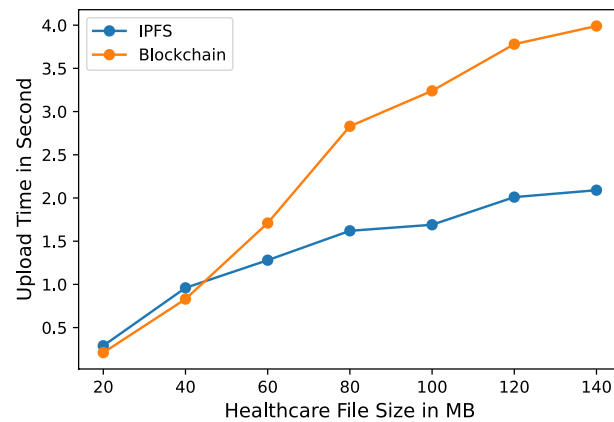


Fig. 5 IPFS vs. Blockchain upload times

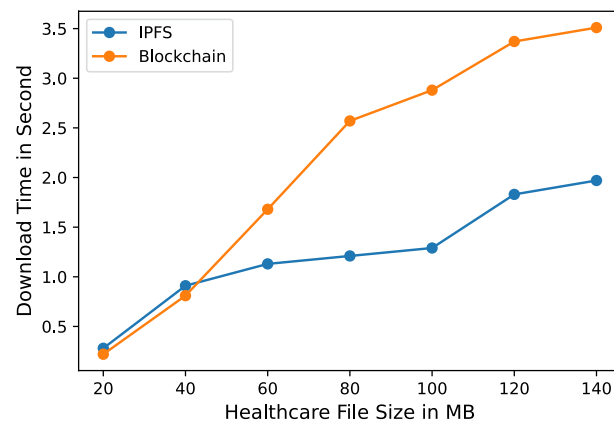


Fig. 6 Blockchain vs. IPFS download times

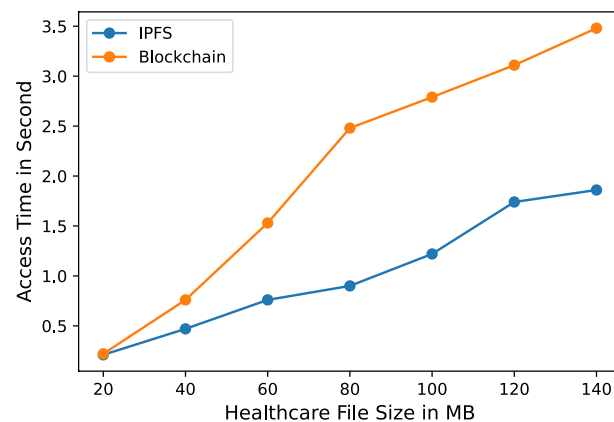
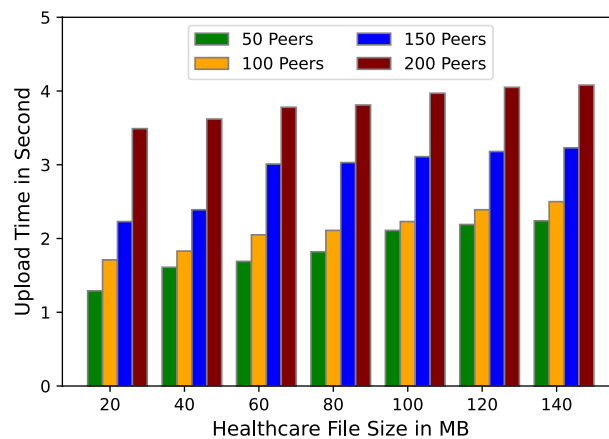


Fig. 7 Blockchain vs. IPFS access times

more time than downloading the same file from IPFS. The IPFS and Blockchain network file access times are depicted in Fig. 7. In an experiment, we accessed files from several workstations with different file sizes. It has been observed that accessing a file using the IPFS network is significantly faster than using the Blockchain network. The experimental findings of testing several file sizes of healthcare data on different machines show that the IPFS network requires much less time for uploading, downloading, and accessing

Table 3 Access, Download, and Upload Times in Blockchain and IPFS

File Size (In MB)	Download Time (in Second)		Upload Time (in Second)		Access Time (in Second)	
	Blockchain	IPFS	Blockchain	IPFS	Blockchain	IPFS
20	0.22	0.28	0.21	0.29	0.22	0.21
40	0.81	0.91	0.83	0.96	0.76	0.47
60	1.68	1.13	1.71	1.28	1.53	0.76
80	2.57	1.21	2.83	1.62	2.48	0.9
100	2.88	1.29	3.24	1.69	2.79	1.22
120	3.37	1.83	3.78	2.01	3.11	1.74
140	3.51	1.97	3.99	2.09	3.48	1.86

**Fig. 8** Upload time analysis in IPFS

files than the Blockchain network. We tested the Blockchain and IPFS networks with files of varying sizes to see how quickly they could be uploaded, downloaded, and accessed. The findings are displayed in Table 3.

To test our framework, we have considered 50, 100, 150, and 200 peers connected to the network and uploaded data of 20MB, 40MB, 60MB, 80MB, 100MB, 120MB, and 140MB. Figure 8. depicts the IPFS network upload time analysis. The experimental result shows that the upload time increased with both the number of peers and the size of the file being uploaded, but that at a certain point, the upload time varies very little even when the number of peers and the size of the file both vary. Once a certain threshold is achieved, every further increase in the file size for uploading results in a small increase in uploading time. IPFS download time analysis is shown in Fig. 9. The result shows that the download time increased with both the file size and the number of peers. The time required to download the file significantly decreased when we increased its size while keeping the number of peers constant. The experimental results show that uploading files to the IPFS network takes longer than downloading. Download and Upload times were measured with a number of peer counts and file sizes for experimental evaluation, as shown in Table 4. Figure 10. analyzes the upload and download times between Blockchain and IPFS. Based on experimental evaluation and analysis, it has been observed that the IPFS network has much faster upload and download speeds than the Blockchain network.

The proposed framework's mining time is calculated using 200 peers with varying file sizes. The result shows that Blockchain networks require more time to mine than

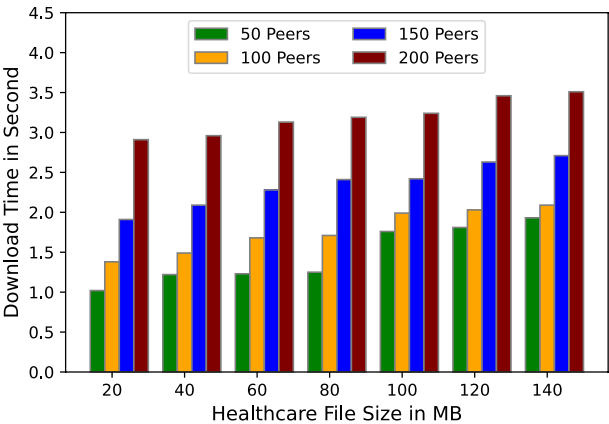


Fig. 9 Download time analysis in IPFS

Table 4 Blockchain and IPFS upload, download, and access times with variable file sizes and peer numbers

File Size (In MB)	Download Time (in Second)				Upload Time (in Second)			
	50 Peers	100 Peers	150 Peers	200 Peers	50 Peers	100 Peers	150 Peers	200 Peers
20	1.02	1.38	1.91	2.91	1.29	1.71	2.23	3.49
40	1.22	1.49	2.09	2.96	1.61	1.83	2.39	3.62
60	1.23	1.68	2.28	3.13	1.69	2.05	3.01	3.78
80	1.25	1.71	2.41	3.19	1.82	2.11	3.03	3.81
100	1.76	1.99	2.42	3.24	2.11	2.23	3.11	3.97
120	1.81	2.03	2.63	3.46	2.19	2.39	3.18	4.05
140	1.93	2.09	2.71	3.51	2.24	2.5	3.23	4.08

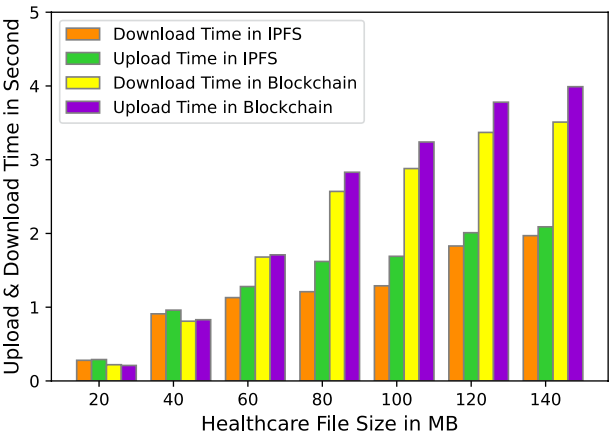


Fig. 10 Blockchain and IPFS upload and download times

IPFS. When comparing Blockchain with IPFS, the mining time in a Blockchain network is slightly greater for small file sizes and steadily increases as file size increases. The mining time analysis for 150 peers is depicted in reference Fig. 11. The comparison of size differences between IPFS and Blockchain ledgers are depicted in reference Fig. 12. Experimental evaluation shows that as more files are uploaded, IPFS storage expands in proportion to the size of the files, while the Blockchain ledger expands at a constant

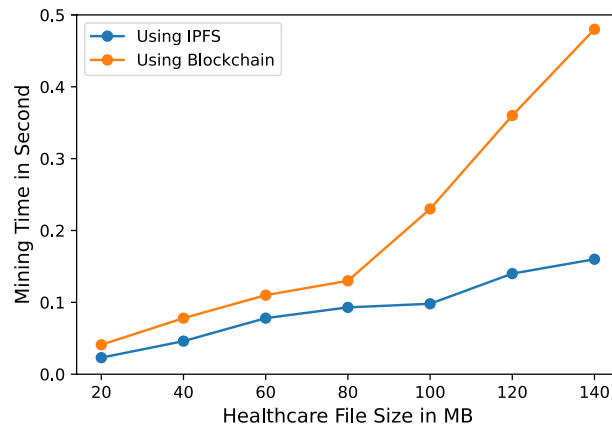


Fig. 11 Blockchain and IPFS mining time comparison with 200 peers

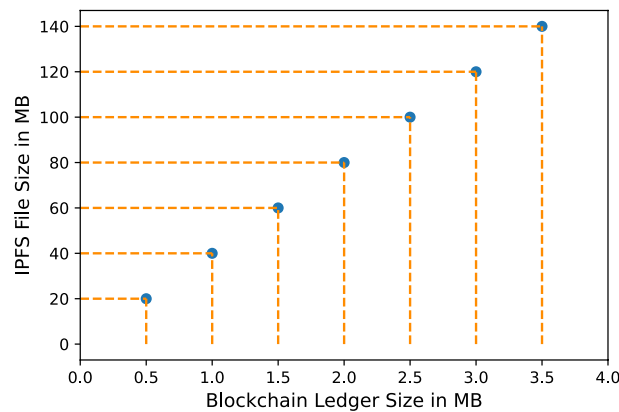


Fig. 12 IPFS file size vs. Blockchain ledger size

rate because only hash values are stored on the Blockchain network, which is a fixed size regardless of file size. Therefore, the Blockchain file sizes increase with hash size.

The proposed blockchain, IoT, and IPFS-integrated healthcare system with decentralised data storage, distributed processing, and off-chain data management is more scalable. IPFS stores real-time health data from IoT devices, reducing blockchain load by only maintaining content hashes. This improves throughput by reducing the amount and frequency of transactions, which in turn minimises congestion on the network. This solution reduces latency and avoids limitations caused by centralised cloud services by transferring large medical files to IPFS. Supporting the growing demand in current healthcare infrastructures, this layered architecture allows for horizontal scalability by adding more devices and users without severely reducing system performance.

6 Findings and discussions

Table 5 compares the proposed framework based on an Interplanetary File System to existing IoT and Blockchain-integrated models regarding security, privacy, scalability, interoperability, efficiency, access management, and vulnerability checks. This research demonstrates that most currently available models run into problems due to technological limitations, are often unable to provide real-time patient care, and experience network overload when processing massive volumes of data and connecting vast numbers

Table 5 Technical comparison of Proposed Framework vs Existing Blockchain-Integrated IoT Models

Contributions	Security	Privacy	Scalability	Interoperability	Efficiency	Access Management	Vulnerability Check-up	Case Study
S. S. Hazari, et al. [32]	✓	✓	X	X	✓	X	X	X
N. Alrebdi, et al. [26]	✓	✓	✓	✓	X	X	X	X
L. Tan, et al. [27]	✓	✓	X	✓	✓	✓	X	X
M. Y. Jabarulla, et al. [28]	✓	✓	✓	✓	X	✓	X	X
J. Sun, et al. [29]	✓	✓	✓	✓	X	✓	X	X
S. Chenthara, et al. [33]	✓	✓	✓	✓	X	X	X	X
Z. Sun, et al. [34]	✓	✓	✓	✓	✓	✓	X	X
A. Jyoti, et al. [35]	✓	✓	✓	✓	✓	X	X	X
Proposed Framework	✓	✓	✓	✓	✓	✓	✓	✓

of devices. Most models do not perform well and do not deliver a better result based on the features we have considered. In contrast to earlier models, the proposed architecture produces superior results for the aforementioned characteristics. In contrast to earlier models, the proposed architecture produces superior results for the aforementioned parameters. Therefore, the suggested framework can be implemented in healthcare institutions to provide confidentiality, integrity, interoperability, and scalability of patient records and device connections.

Integration of blockchain technology, the IoT, and immutable public file systems (IPFS) improves healthcare delivery through decentralising data management, providing immutable medical records, and protecting device connections. Data manipulation and unauthorised access are reduced by the decentralised nature of blockchain, which mitigates single points of failure. Blockchain's immutable ledger and smart contract-based access control help IoT devices protect sensitive health data against lateral movement threats. Because IPFS distributes encrypted data over multiple nodes, it becomes more difficult to carry out a breach, significantly strengthening security. To address issues raised by HIPAA and GDPR, further levels of protection can be added using cryptographic hashing algorithms and proxy re-encryption. Although consensus mechanisms in blockchain make processing slower, off-chain storage using IPFS makes the system more scalable. Within this framework, smart contracts are able to implement regulatory logic in a way that is both transparent and protects patients' privacy.

6.1 Use case

Chronic disease remote monitoring: A patient with a chronic illness, such as diabetes, employs IoT gadgets, including a glucose monitor and smartwatch, to track their health. Medical professionals have access to the data in real time and can analyse and make decisions based on it because it is stored securely, which is depicted in figure Fig. 13.

1. **Data Collection:** IoT devices collect sensitive patient data like glucose levels, heart rate, etc.
2. **Data Encryption:** Data is encrypted using the provided encryption technique.
3. **IPFS Data Storage:** The IPFS stores the encrypted data and generates the unique CID. The blockchain keeps track of transactions.
4. **Blockchain Storage:** Metadata, including the patient's ID and the timestamp along with the CID, are recorded on the blockchain log.
5. **Data Access:** The CID allows healthcare providers to access, decrypt, and analyse data for diagnosis or treatment, reducing the need for healthcare visits.

7 Conclusions

The experimental findings show that the proposed framework outperforms existing models across various dimensions. The architecture improves security, privacy, interoperability, scalability, and overall efficiency for healthcare stakeholders and patients by integrating blockchain technology with IoT systems and IPFS. IPFS efficiently handles storage, access, and scalability issues, while the blockchain exclusively stores cryptographic hashes, leaving minimal system overhead. A device proxy is also used to access and monitor unprotected devices securely. The system's performance was extensively evaluated with various file sizes and peer counts. The results demonstrate that the proposed framework allows much faster upload, download, and access times compared to

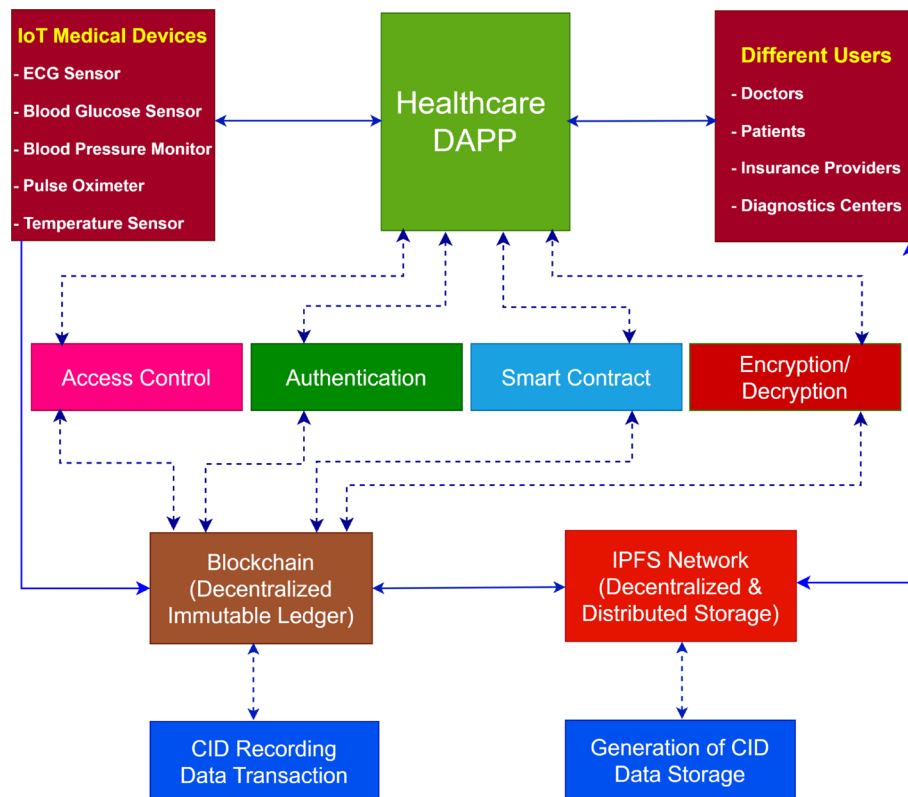


Fig. 13 Use case diagram

traditional blockchain-based healthcare systems. A key component of the system's speed and lightweight nature is the blockchain ledger, which grows consistently, unlike IPFS storage, which grows according to file size. The major advantages of the proposed decentralized healthcare architecture over conventional centralized systems are superior efficiency, increased storage capacity, faster data access, and the removal of single points of failure.

To further improve data security in Blockchain-enabled healthcare, future studies can investigate the incorporation of cryptographic algorithms that are resistant to quantum computing. Predictive analytics powered by artificial intelligence used to datasets protected by blockchain technology can facilitate early diagnosis and individualised treatment plans.

Author contributions

S. R. Mallick performed the conceptualisation, methodology, formal analysis, and original draft writing. Supervision, review, and editing were conducted by S. Sobhanayak and R. K. Lenka.

Funding

Open access funding provided by Symbiosis International (Deemed University).

Data availability

All data generated or analysed during this study are included in this published article.

Code availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no Conflict of interest.

Received: 12 May 2025 / Accepted: 29 July 2025

Published online: 19 August 2025

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